CBSE – CLASS XII PRACTICE QUESTION PAPER

Subject: PHYSICS
Chapter: Ray Optics and Optical Instruments
Time Allowed: 1 Hour Maximum Marks: 30

General Instructions:

1. All questions are compulsory.
2. The paper has 10 questions in three sections: Section A (Q1–4, 2 marks each), Section B (Q5–8, 3 marks each) and Section C (Q9–10, 5 marks each).
3. Show all steps in numerical problems with proper units. Draw neat, labelled diagrams wherever asked.

SECTION A (Q1–Q4) – 2 marks each

Q1. Draw a labelled ray diagram to show the image formation by a concave mirror when the object is placed between its pole and principal focus. State the nature of the image formed. **[2]**

Q2. Light travels from glass of refractive index 1.5 into air. Calculate (i) the critical angle for the glass–air interface and (ii) the speed of light in glass. (Take $c = 3 \times 10^8$ m s^{-1}, $\sin 41.8° \approx 0.667$) **[2]**

Q3. Define total internal reflection. State two conditions necessary for it to occur and give one practical application. **[2]**

Q4. An object 2 cm high is placed 15 cm in front of a concave mirror of focal length 10 cm. Find the position, nature and size of the image formed. **[2]**

SECTION B (Q5–Q8) – 3 marks each

Q5. Draw a labelled ray diagram for the refraction of a ray of light through an equilateral glass prism, showing the angle of deviation. Hence, deduce the expression for the refractive index of the prism material in terms of the angle of the prism (A) and the angle of minimum deviation (δ_m). **[3]**

Q6. A double convex lens is made of glass of refractive index 1.5, with both faces of radius of curvature 20 cm. Find (i) its focal length and power in air, and (ii) its focal length when it is completely immersed in water of refractive index 4/3. **[3]**

Q7. Draw a labelled ray diagram of a compound microscope when the final image is formed at the least distance of distinct vision. Write the expression for its magnifying power. **[3]**

Q8. An astronomical telescope has an objective of focal length 100 cm and an eyepiece of focal length 5 cm. Find (i) its magnifying power and tube length in normal adjustment, and (ii) its magnifying power and tube length when the final image is formed at the least distance of distinct vision (25 cm). **[3]**

SECTION C (Q9–Q10) – 5 marks each

Q9. Draw a labelled diagram to show the refraction of light at a convex spherical surface separating two media of refractive indices n_1 and n_2 ($n_2 > n_1$) when a point object is placed in the rarer medium. Using the figure, derive the relation between the object distance u, image distance v and radius of curvature R: $n_2/v - n_1/u = (n_2 - n_1)/R$. State the assumptions made. **[5]**

Q10. A convex lens of focal length 30 cm is placed in contact with a concave lens of focal length 50 cm. (a) Find the power and the focal length of the combination. (b) An object 5 cm high is placed 150 cm in front of this combination. Find the position, nature and size of the image. **[5]**